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Kumar

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- (54) **HYBRID LASER AND LED BASED ILLUMINATION DEVICE**

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F21Y 113/20 (2016.01)
F21Y 115/10 (2016.01)
F21Y 115/30 (2016.01)
H01L 25/075 (2006.01)

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(2015.01); *H01L 25/0753* (2013.01); *F21Y*
2105/16 (2016.08); *F21Y 2113/20* (2016.08);
F21Y 2115/10 (2016.08); *F21Y 2115/30*
(2016.08)

- (58) **Field of Classification Search**
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See application file for complete search history.

- (56) **References Cited**

U.S. PATENT DOCUMENTS

9,853,413	B2 *	12/2017	Kim	H01S 5/02253
10,263,395	B2 *	4/2019	Kim	H01S 5/423

- | | | | | |
|--------------|------|---------|--------------------|----------------------------|
| 2004/0217364 | A1 * | 11/2004 | Tarsa | H01L 25/0753
257/89 |
| 2012/0229292 | A1 * | 9/2012 | Laabs | F21V 29/507
340/815.45 |
| 2013/0121759 | A1 * | 5/2013 | Breidenassel | F21V 29/50
403/330 |
| 2016/0131318 | A1 * | 5/2016 | Kim | H01S 5/02253
362/311.02 |
| 2016/0165680 | A1 * | 6/2016 | Johnson | F21V 23/005
438/23 |
| 2018/0159303 | A1 * | 6/2018 | Kim | F21V 5/043 |
| 2024/0170911 | A1 * | 5/2024 | Templier | H01L 25/0753 |

FOREIGN PATENT DOCUMENTS

- | | | | | | | |
|----|---------------|----|---|--------|-------|------------|
| CN | 102918930 | A | * | 2/2013 | | F21S 8/006 |
| KR | 20080062490 | A | * | 7/2008 | | |
| WO | WO-2024163722 | A2 | * | 8/2024 | | A45F 3/005 |

* cited by examiner

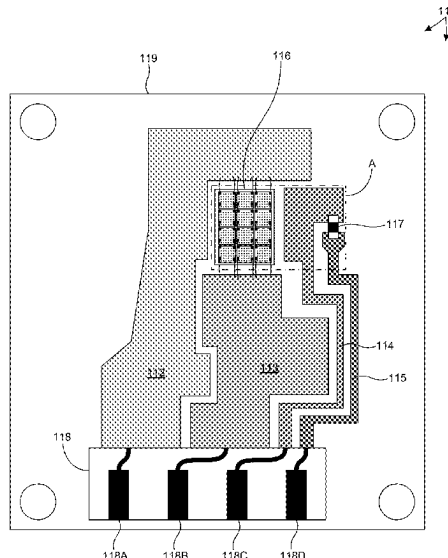
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Joseph S. Spano

(57) **ABSTRACT**

Methods and systems for combining electrically isolated LEDs and one or more laser diodes into a hybrid illumination device are presented herein. A hybrid LED and laser diode based illumination device includes both electrically isolated LEDs and one or more laser diodes assembled on common metal core printed circuit board. Employing electrically isolated LEDs allows the electrical connection of multiple LEDs in series on the common metal core printed circuit board with high spatial density. In some embodiments, a laser diode source is located within a spatial envelope of an LED array to realize a particularly compact high optical power, hybrid LED and laser diode based illumination device with a relatively small illumination source etendue. In a further aspect, different sets of LEDs and one or more laser diodes are independently controlled to generate illumination with different spectral characteristics.

20 Claims, 9 Drawing Sheets



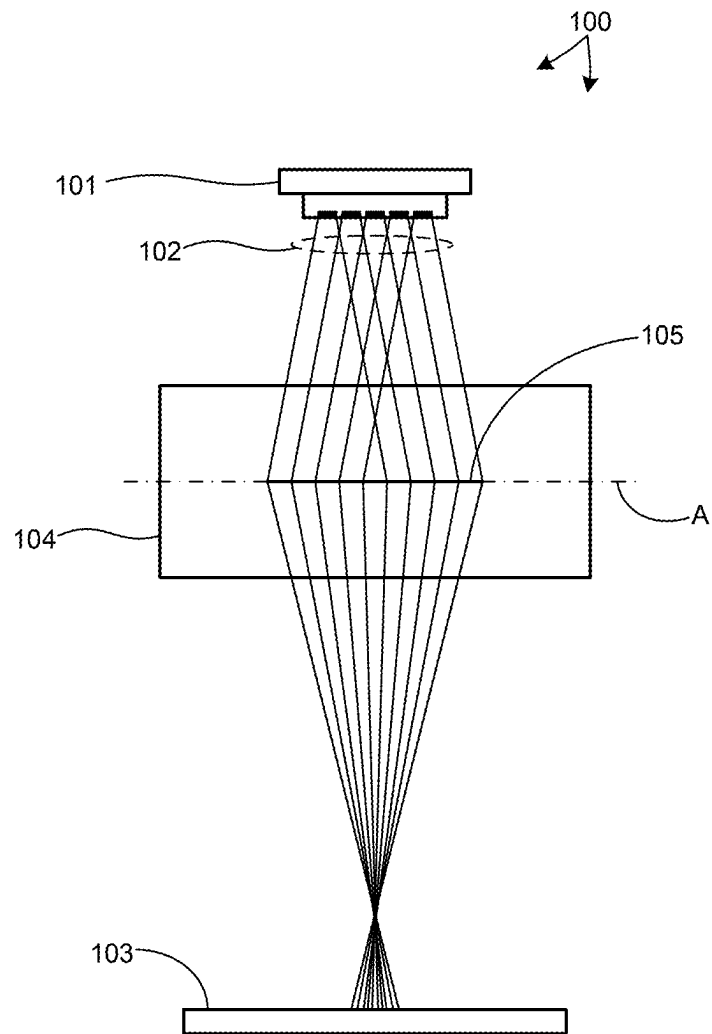


FIG. 1

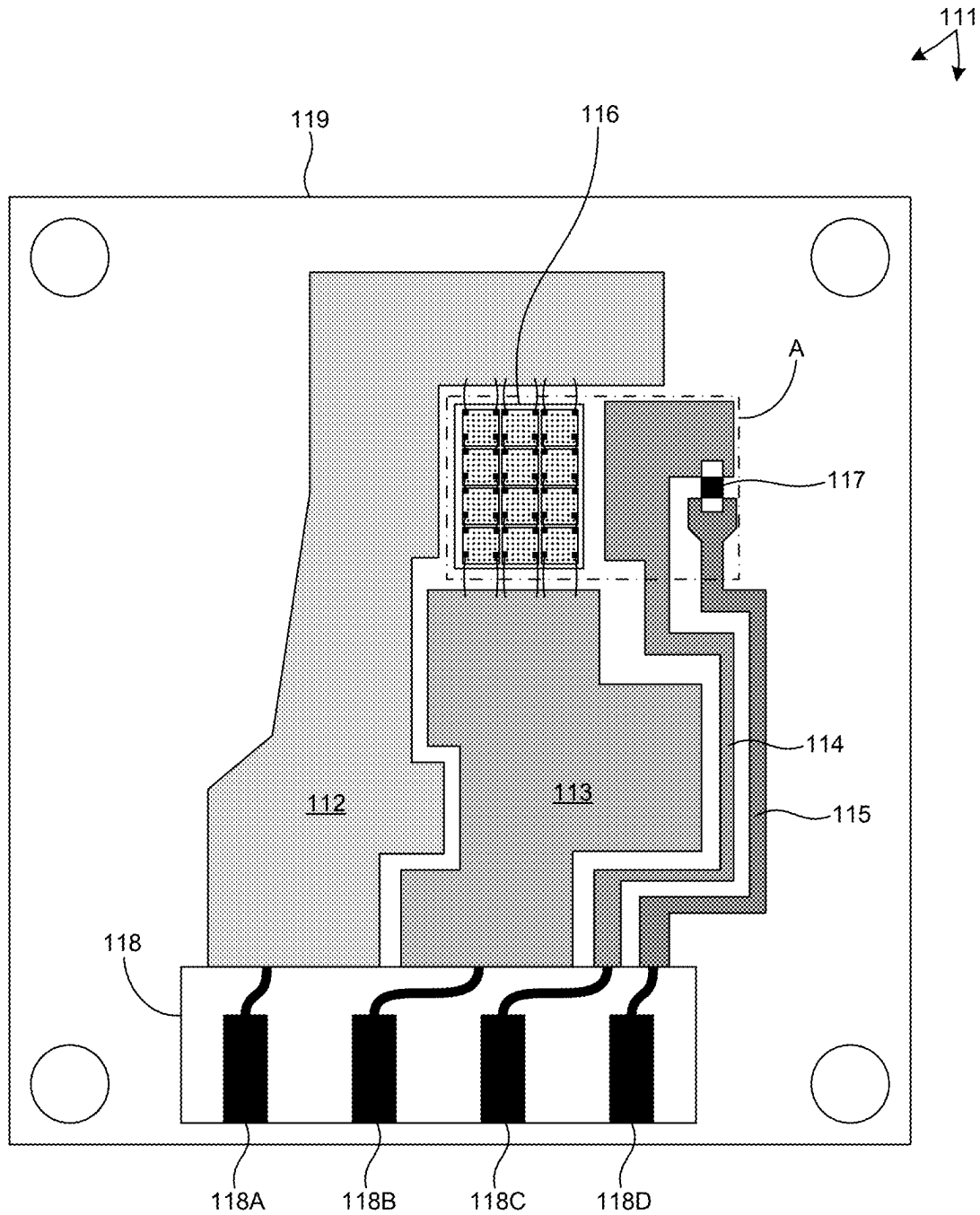


FIG. 2

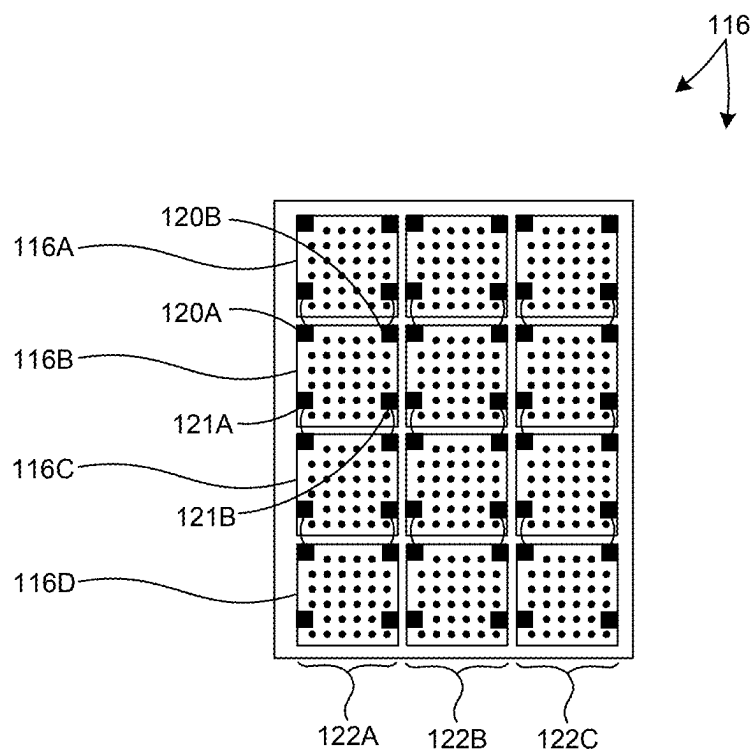


FIG. 3

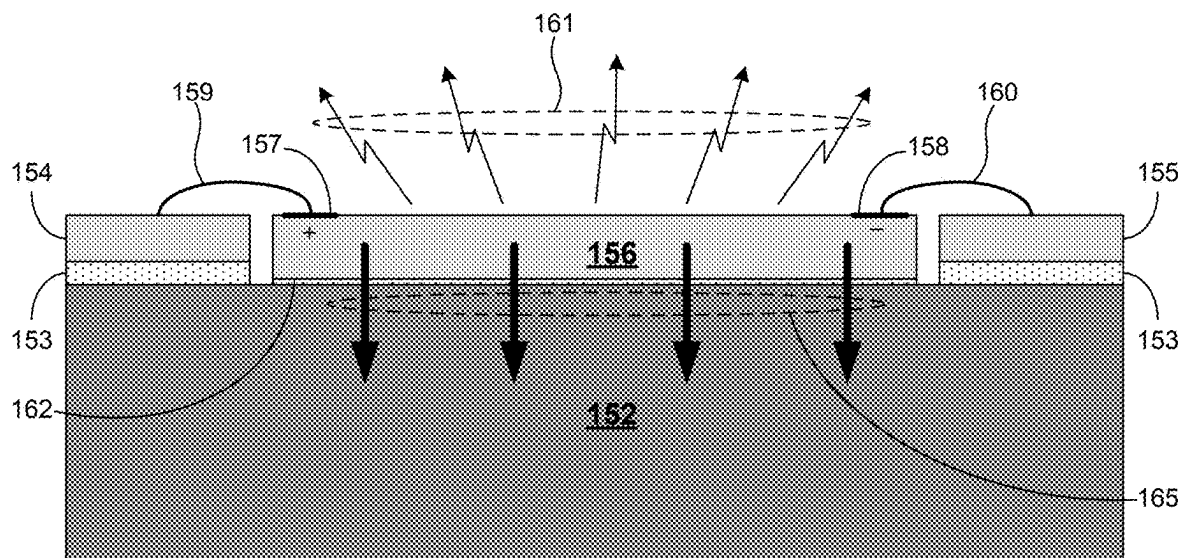


FIG. 4

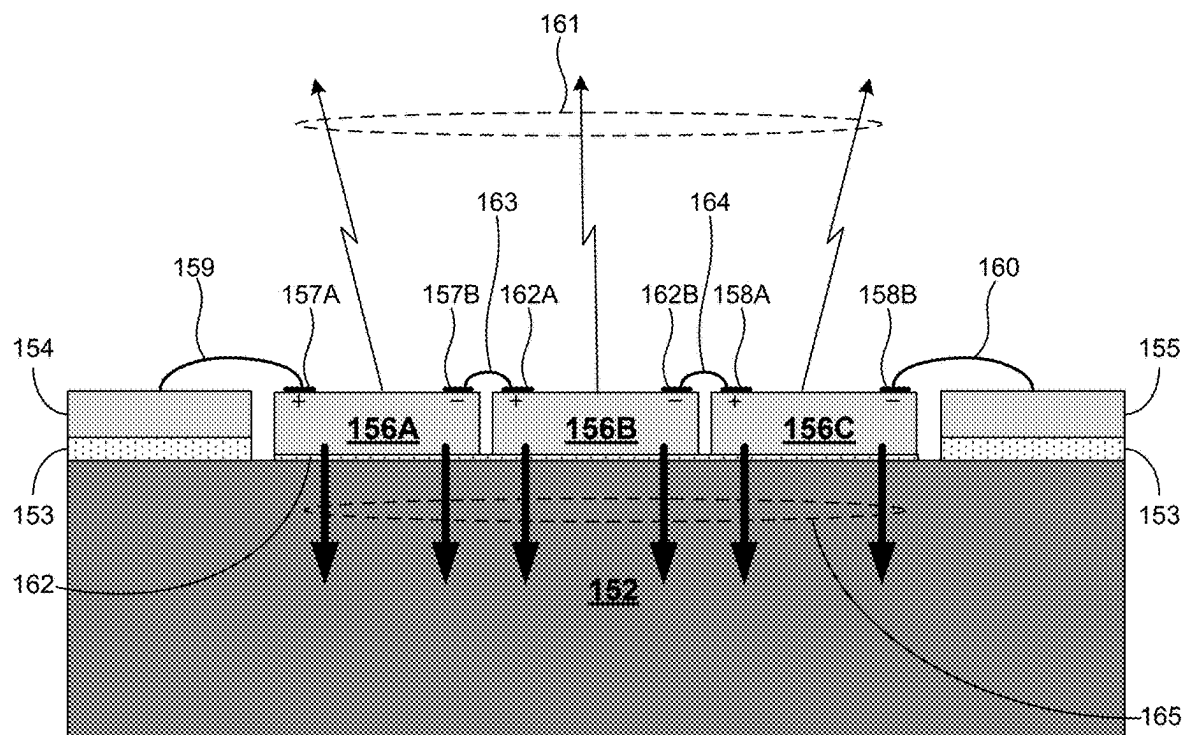


FIG. 5

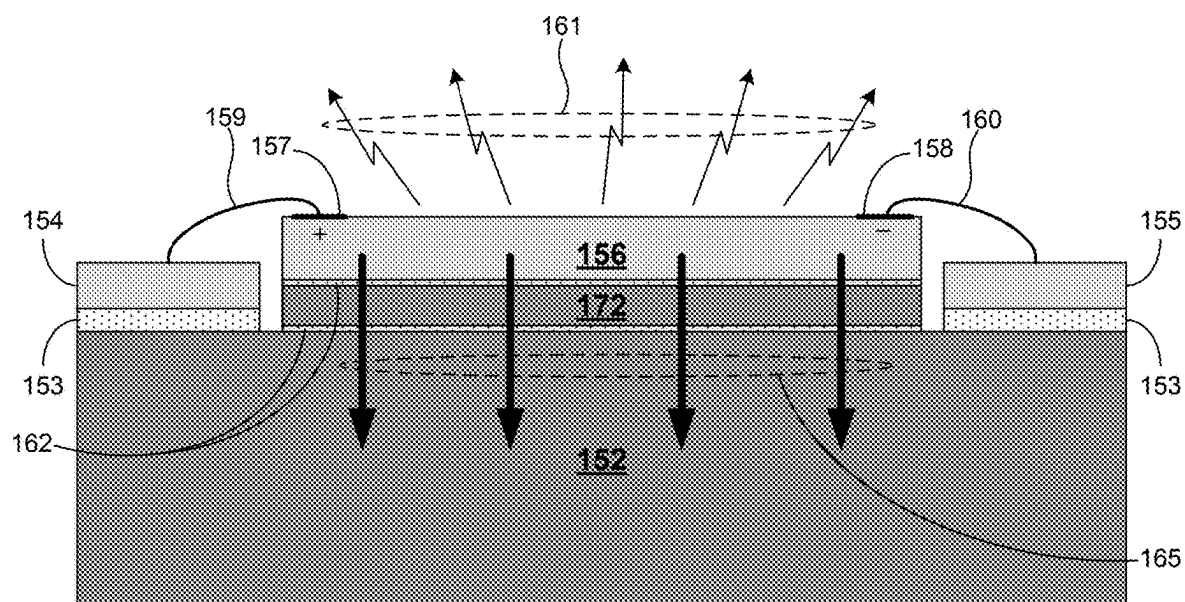


FIG. 6

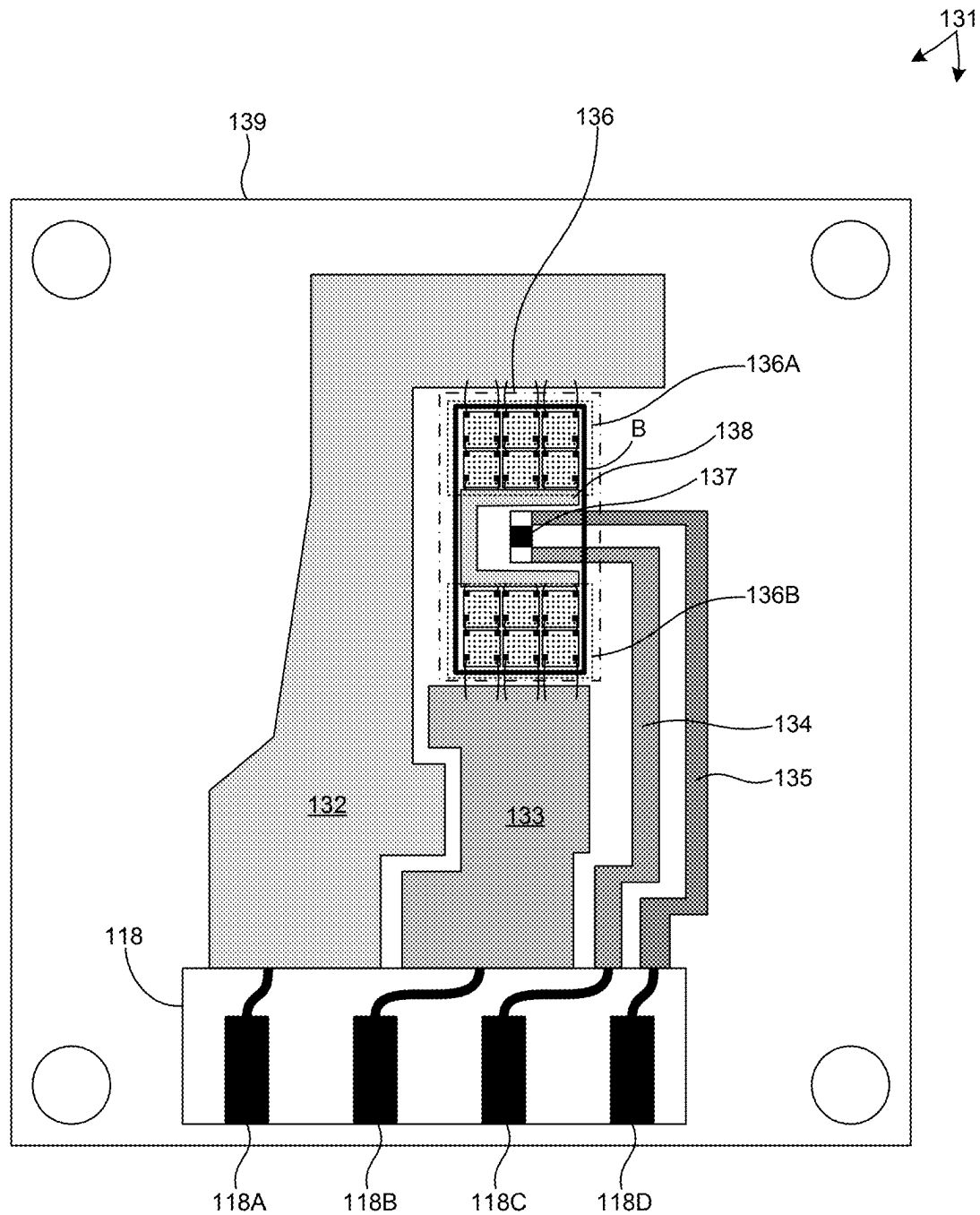


FIG. 7

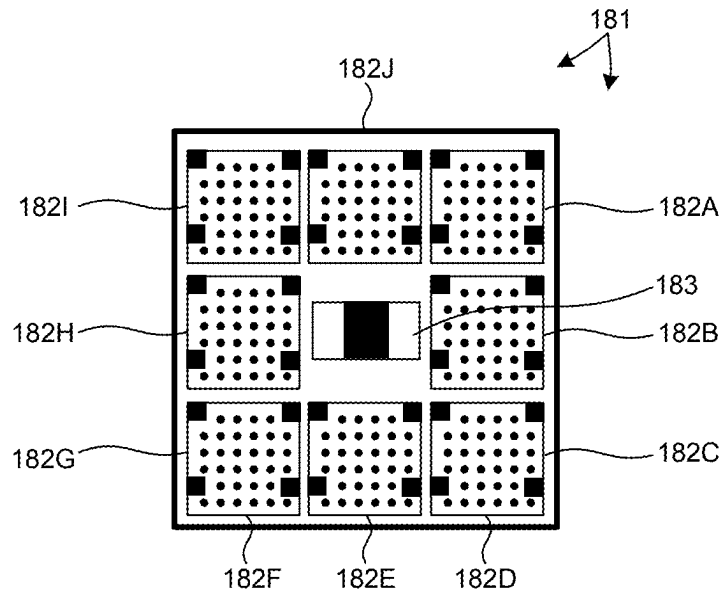


FIG. 8

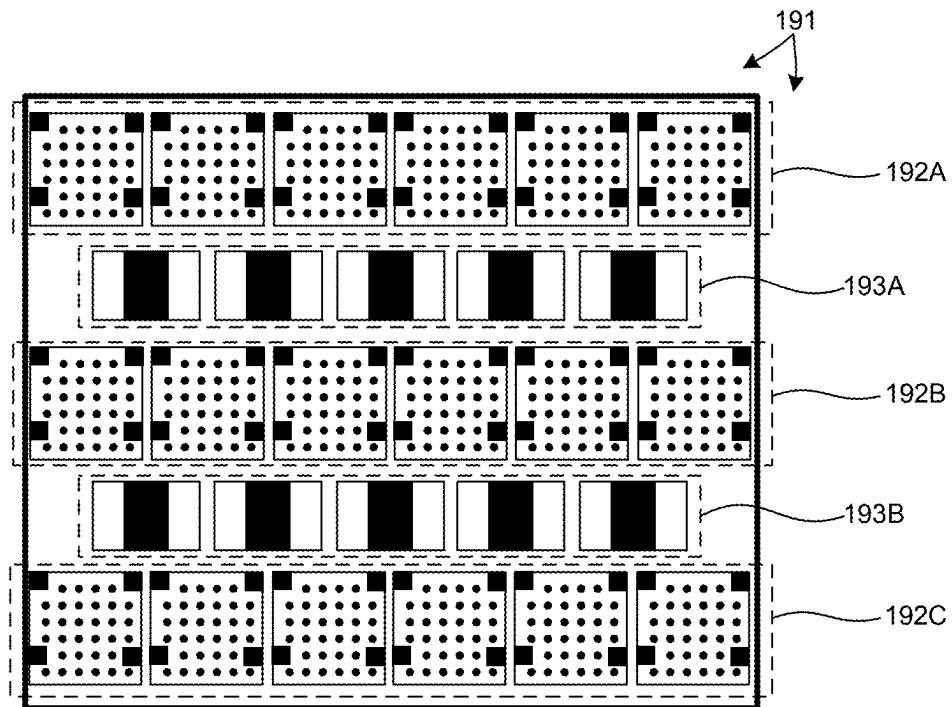


FIG. 9

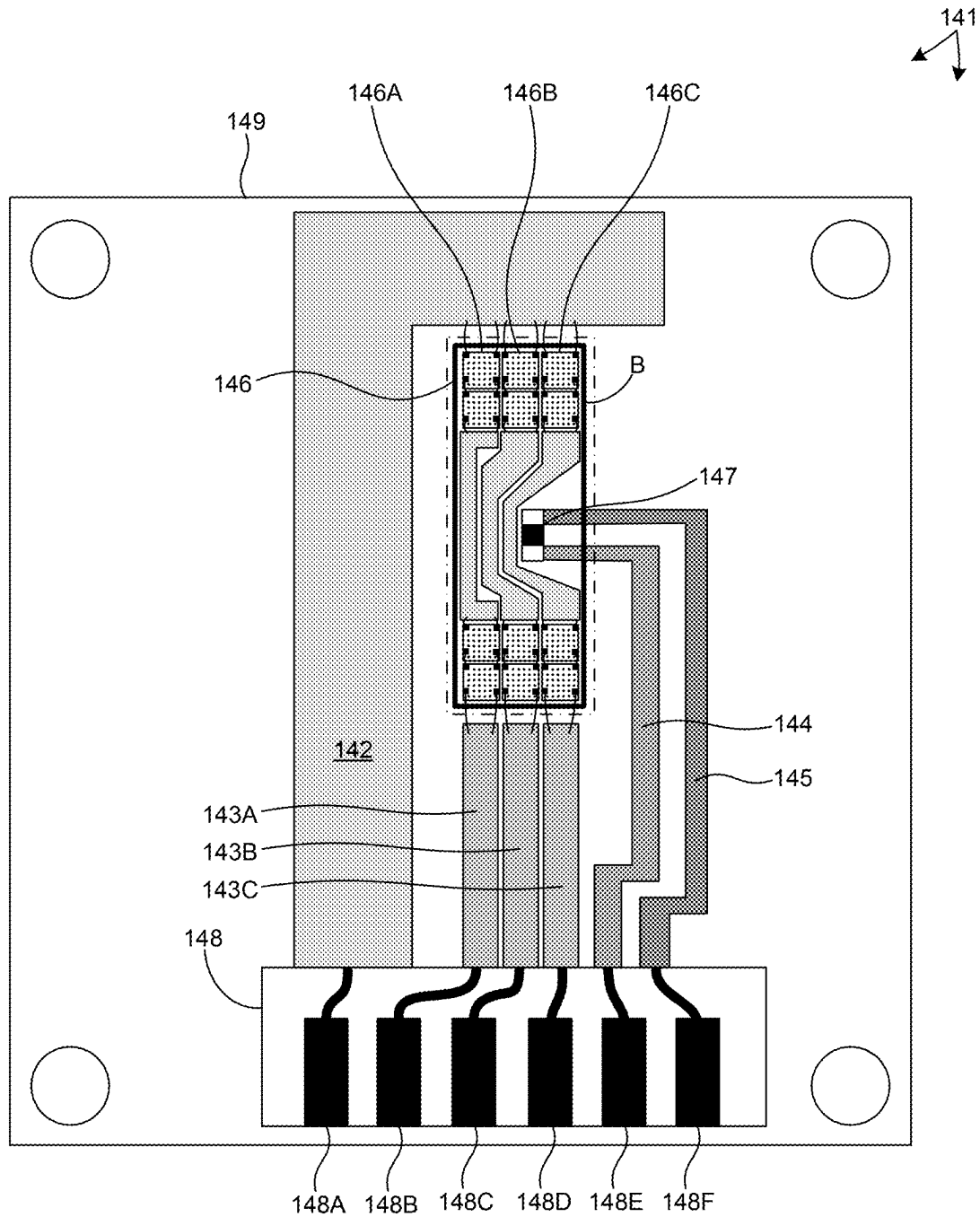


FIG. 10

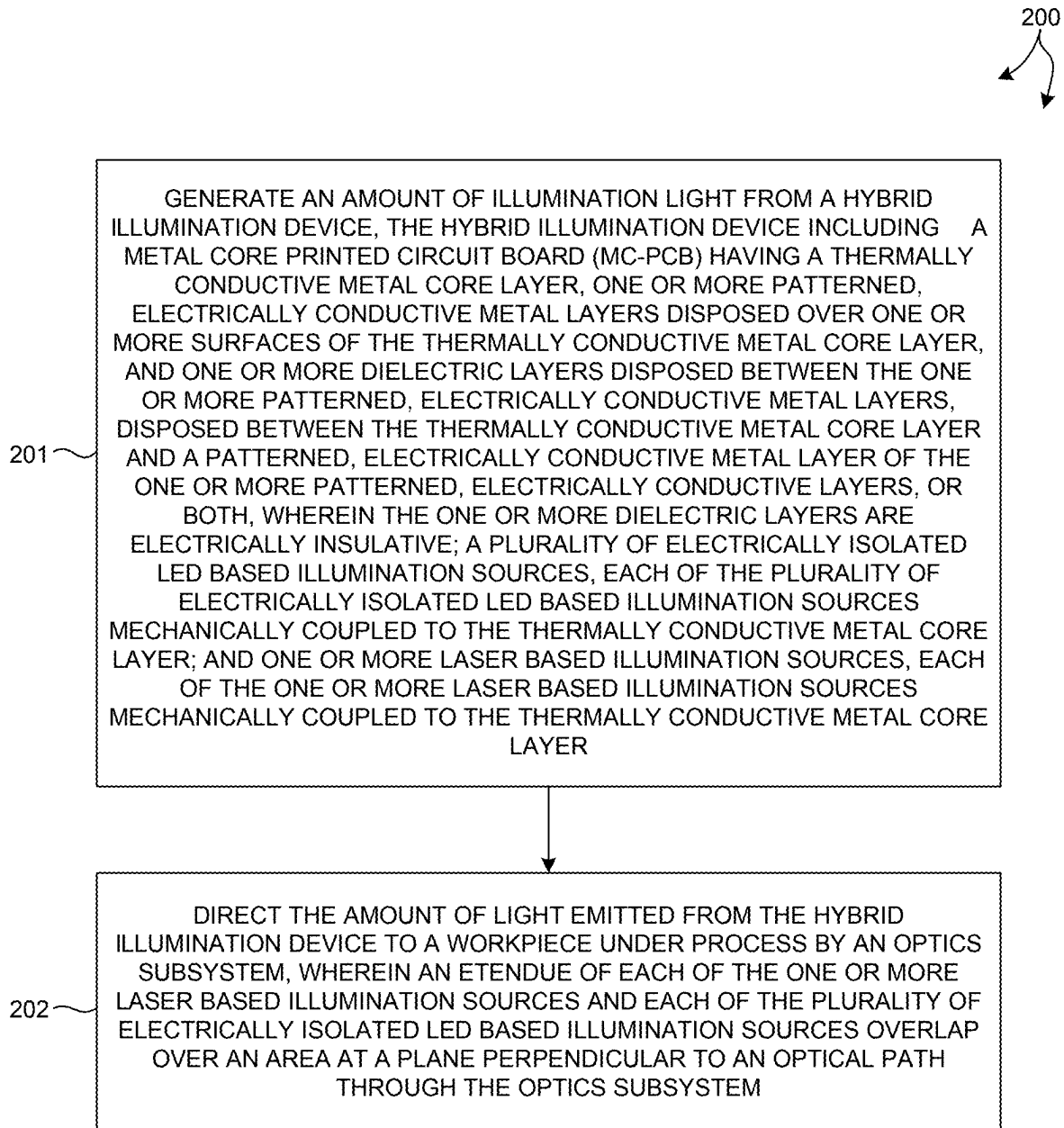


FIG. 11

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**HYBRID LASER AND LED BASED
ILLUMINATION DEVICE****TECHNICAL FIELD**

The described embodiments relate to illumination device that include a Laser diode based illumination source and a Light Emitting Diode (LED) based illumination source.

BACKGROUND INFORMATION

Light emitting devices employing Light Emitting Diode (LED) or Laser Diode (LD) illumination sources have achieved broad market penetration due to their electrical efficiency, long lifetime, high reliability, relatively low cost of manufacture, and physical scalability. Markets addressed by LED based light emitting devices include vehicle lighting, indoor and outdoor building lighting, display lighting, etc. In particular, LED based light emitting devices are rapidly replacing traditional illumination sources such as incandescent bulbs and fluorescent bulbs in general illumination applications.

In many industrial applications LEDs and LDs are employed at ultraviolet wavelengths to cure different materials, e.g., adhesives, etc. In these applications, relatively high optical power is required to meet throughput requirements of high volume processing equipment, e.g., semiconductor packaging applications. Industrial processing equipment is employed to cure many different process materials, often at different wavelength ranges. Unfortunately, as narrowband emitters, LEDs and LDs are not directly suited to illumination applications that require high power illumination at a number of different spectral bands.

In some examples, one or more different illumination sources are selected to illuminate the workpiece under process as required by the process application. In some examples, the output of each different illumination source is independently controlled, e.g., switched on/off. In some examples, the output of each different illumination source is selectively directed to the workpiece under process or away from the workpiece under process by movement of an optical element of the optics system, e.g., a flip-in mirror, mechanical shutter, etc.

In some examples, several different illumination sources, each emitting at different spectral bands, are employed. However, the etendue of one or more of the different illumination sources do not overlap. Thus, the optical path from one or more of the different illumination sources is not the same as other illumination sources. To overcome this limitation, the optical output of different illumination sources is optically combined using various discrete optical elements, e.g., mirrors, beam splitters, etc., located or selectively located in an optical path specific to a subset of the different illumination sources. In these examples, one or more optical elements do not interact with light output from all of the different illumination sources. This approach increases the cost and complexity of the optical system.

Current process equipment requires separate illumination optics associated with different illumination sources to effectively combine illumination light required to process different materials at different spectral ranges. This approach complicates the optical design of the process equipment, introduces optical losses, and increases equipment cost.

Future process applications present challenges for high throughput curing of different materials at high optical

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power and different spectral ranges. Thus, methods and systems for improved illumination power and spectral control are desired.

SUMMARY

Methods and systems for combining electrically isolated LEDs and one or more laser diodes into a hybrid illumination device are presented herein. A hybrid LED and laser diode based illumination device includes both electrically isolated LEDs and one or more laser diodes assembled on common metal core printed circuit board. Employing electrically isolated LEDs allows the electrical connection of multiple LEDs in series on the common metal core printed circuit board. This enables the integration of multiple LEDs and one or more laser diodes with high spatial density.

In some embodiments, illumination generated by a hybrid illumination device is directed to an optics subsystem. The etendue of each of the illumination sources of the hybrid illumination device overlap over an area perpendicular to an optical path through the optics subsystem. Furthermore, each of the illumination sources share a common optical path through the optics subsystem from the hybrid illumination device to a workpiece under process. In this manner, the optical output of any combination of the illumination sources of the hybrid illumination device is projected onto the workpiece under process using the same optical elements in the same configuration.

In one aspect, a hybrid LED and laser diode based illumination device includes both electrically isolated LEDs and one or more laser diodes assembled on a common metal core printed circuit board. In preferred embodiments, each of the one or more laser diode illumination sources is also electrically isolated.

Electrically isolated LEDs flow current through top facing terminals rather than the backside surface of the LED die. In some embodiments, multiple bare LED die are mounted to a common MC-PCB and wired in series using wire bonds on top of the LED die. In high optical power applications, it is advantageous to couple multiple bare LED die in series to minimize the current that must be generated by a current driver to drive multiple LED die. Furthermore, it is advantageous to couple multiple bare LED die in series to maximize the spatial density of LED die mounted to a MC-PCB. This minimizes the illumination source etendue and simplifies the optics subsystem required to focus the illumination light on the workpiece.

In some embodiments, a laser diode source is located within a spatial envelope of an LED array to realize a particularly compact, high optical power, hybrid LED and laser diode based illumination device. By locating the laser diode source within the spatial envelope of the LED array, the illumination source etendue of the hybrid LED and laser diode based illumination device may be further reduced compared to a design where the laser diode source is located outside the spatial envelope of the LED array.

In some embodiments, a hybrid LED and laser diode based illumination device includes a multiple layer MC-PCB. In some of these embodiments, the electrical traces electrically coupled to one or more laser diode based illumination sources are patterned from a different electrically conductive metal layer than the electrical traces electrically coupled to one or more LED based illumination sources of the hybrid LED and laser diode based illumination device.

In a further aspect, different sets of LEDs and one or more laser diodes are independently controlled to generate illumination with different spectral characteristics. In this man-

ner, the illumination characteristics of the hybrid LED and laser diode based illumination device are adapted to applications that require different combinations of illumination wavelengths. In some examples, different combinations of illumination sources of the hybrid LED and laser diode based illumination device are employed using the same optical subsystem to perform industrial curing of a wide range of materials.

The foregoing is a summary and thus contains, by necessity, simplifications, generalizations and omissions of detail; consequently, those skilled in the art will appreciate that the summary is illustrative only and is not limiting in any way. Other aspects, inventive features, and advantages of the devices and/or processes described herein will become apparent in the non-limiting detailed description set forth herein.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a simplified diagram illustrative of an illumination system including a hybrid LED and laser diode based illumination device and an optics subsystem.

FIG. 2 is a simplified diagram illustrative of a hybrid LED and laser diode based illumination device in one embodiment including an array of electrically isolated LEDs and a laser diode mounted to a metal core printed circuit board (MC-PCB).

FIG. 3 is a simplified diagram illustrative of an LED array including three sets of electrically isolated, bare LED die, each including four electrically isolated, bare LED die wired in series.

FIG. 4 is a simplified diagram illustrative of a cross-sectional view of an electrically isolated LED based illumination source assembled onto a single layer MC-PCB in a wire bond configuration.

FIG. 5 is a simplified diagram illustrative of a cross-sectional view of multiple electrically isolated LED based illumination sources assembled onto a single layer MC-PCB and electrically coupled in series in a wire bond configuration.

FIG. 6 is a simplified diagram illustrative of a cross-sectional view of an electrically isolated LED based illumination source assembled onto a single layer MC-PCB with a pedestal structure.

FIG. 7 is a simplified diagram illustrative of a hybrid LED and laser diode based illumination device in another embodiment.

FIG. 8 is a simplified diagram illustrative of a hybrid LED and laser diode based illumination device in another embodiment.

FIG. 9 is a simplified diagram illustrative of a hybrid LED and laser diode based illumination device in another embodiment.

FIG. 10 is a simplified diagram illustrative of a hybrid LED and laser diode based illumination device in another embodiment.

FIG. 11 is a flowchart illustrative of a method for generating illumination light from a hybrid LED and laser diode based illumination device of the present invention.

DETAILED DESCRIPTION

Reference will now be made in detail to background examples and some embodiments of the invention, examples of which are illustrated in the accompanying drawings.

Methods and systems for combining electrically isolated LEDs and one or more laser diodes into a hybrid illumina-

tion device are presented herein. A hybrid LED and laser diode based illumination device includes both electrically isolated LEDs and one or more laser diodes assembled on common metal core printed circuit board. Employing electrically isolated LEDs allows the electrical connection of multiple LEDs in series on the common metal core printed circuit board. This enables the integration of multiple LEDs and one or more laser diodes with high spatial density.

FIG. 1 depicts a simplified illustration of an illumination system **100** including a hybrid LED and laser diode based illumination device **101** and an optics subsystem **104**. As depicted in FIG. 1, illumination light **102** generated by hybrid LED and laser diode based illumination device **101** is collected by optics subsystem **104** and collimated or focused onto workpiece **103**. As depicted in FIG. 1, the etendue of each of the illumination sources of the hybrid illumination device overlap over an area **105** of plane A. Plane A is a plane perpendicular to an optical path through the optics subsystem **104**. Furthermore, each of the illumination sources share a common optical path through optics subsystem **104** from the hybrid illumination device **101** to the workpiece **103** under process. In this manner, the optical output of any combination of the illumination sources of the hybrid illumination device is projected onto the workpiece under process using the same optical elements in the same configuration. In this manner, the hybrid illumination device is able to illuminate a workpiece under process, e.g., an industrial curing process, using any combination of illumination sources using the same optical elements, without reconfiguring the optical system. In some examples, optical subsystem **104** includes a single focusing optical element, e.g., a refractive objective. In some other examples, multiple optical elements, e.g. mirrors, may be employed to collect illumination light **102** and collimate or focus illumination light **102** onto workpiece **103**.

In one aspect, a hybrid LED and laser diode based illumination device includes both electrically isolated LEDs and one or more laser diodes assembled on a common metal core printed circuit board. FIG. 2 depicts a simplified illustration of a hybrid LED and laser diode based illumination device **111** including an array **116** of electrically isolated LEDs and a laser diode **117** mounted to a metal core printed circuit board (MC-PCB) **119**. In preferred embodiments, each of the one or more laser diode illumination sources is also electrically isolated, i.e., the surface of the laser diode die or package mounted to the MC-PCB is not electrically active.

In general, a MC-PCB includes a thermally conductive metal core layer and at least one patterned, electrically conductive metal layer disposed over one or both sides of the thermally conductive metal core layer. The thermally conductive metal core layer is a relatively thick layer of thermally conductive material, e.g., copper, aluminum, etc. A patterned, electrically conductive metal layer is a relatively thin layer of electrically conductive material or combination of materials, e.g., copper, tin, silver, gold, etc.

In the embodiment depicted in FIG. 2, MC-PCB **119** is a single layer PCB. As depicted in FIG. 2, electrical traces **112-115** are patterned from a single electrically conductive metal layer disposed over a thermally conductive metal core layer. In a single layer MC-PCB, a dielectric layer is disposed between the patterned, electrically conductive metal layer and the thermally conductive metal core layer. The dielectric material is electrically insulative and electrically isolates the patterned, electrically conductive metal layer from the thermally conductive metal core layer.

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As depicted in FIG. 2, portions of the electrically conductive metal layer are selectively etched in a specified pattern. In the example depicted in FIG. 2, the etch pattern creates gaps in the electrically conductive metal layer that electrically isolate electrical traces **112-115** from one another. As depicted in FIG. 2, a multi-terminal connector **118** is mounted to MC-PCB **119**. Terminal connector **118** includes four terminals **118A-D**, electrically coupled to electrical traces **112-115**, respectively. In this manner, electrical terminal **118A** is electrically coupled to one voltage node of LED array **116**, electrical terminal **118B** is electrically coupled to another voltage node of LED array **116**, electrical terminal **118C** is electrically coupled to one voltage node of laser diode **117**, and electrical terminal **118D** is electrically coupled to another voltage node of laser diode **117**.

In the embodiment depicted in FIG. 3, LED array **116** includes three sets of electrically isolated, bare LED die **122A-122C**, each including four electrically isolated, bare LED die, e.g., bare LED die **116A-D**, wired in series. By way of example, bare LED die **116B** includes nodes **121A** and **121B** electrically coupled to the anode of bare LED die **116B** and nodes **120A** and **120B** electrically coupled to the cathode of bare LED die **116B**. As depicted in FIG. 3, the cathode of bare LED die **116D** is wired to the anode of bare LED die **116C**, the cathode of bare LED die **116C** is wired to the anode of bare LED die **116B**, and the cathode of bare LED die **116B** is wired to the anode of bare LED die **116A**. Sets **122B** and **122C** are wired similarly. Furthermore, as depicted in FIG. 2, the anode of bare LED die **116A** is wired to electrical trace **113**, along with the anode of the corresponding bare LED die of sets **122B** and **122C**. Similarly, the cathode of LED bare die **116D** is wired to electrical trace **112**, along with the cathodes of the corresponding bare LED die of sets **122B** and **122C**. In this manner, the three sets of electrically isolated, bare LED die **122A-122C** are electrically coupled in parallel.

In some examples, the sum of the current flow through the three sets of electrically isolated, bare LED die **122A-122C** is controlled by a current driver coupled between terminals **118A** and **118B**. In this manner, the current driver independently controls current flow through LED array **116**. Similarly, another current driver (not shown) is electrically coupled across terminals **118C** and **118D**. This current driver independently controls current flow through laser diode **117**. In this manner, light generated by LED array **116** and laser diode **117** is independently controlled by two different current drivers.

FIG. 4 is a simplified diagram illustrative of a cross-sectional view of an electrically isolated LED based illumination source **156** assembled onto a single layer MC-PCB in a wire bond configuration. As depicted in FIG. 4, the electrically isolated backside of LED based illumination source **156** is disposed over thermally conductive metal core layer **152**. As depicted in FIG. 4, heat flow **165** carries thermal energy away from LED based illumination source **156** into thermally conductive metal core layer **152**. A thin layer of thermally conductive paste **162** is located between LED based illumination source **156** and thermally conductive metal core layer **152**. Thermally conductive paste **162** has high thermal conductivity and fills in any mechanical imperfections between the surfaces of LED based illumination source **156** thermally conductive metal core layer **152** facing one another. This enables a thermal conduction path from the LED based illumination source **156** to the thermally conductive metal core layer **152** with a minimum of thermal resistance. In some examples, the thermally con-

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ductive paste **162** includes a matrix of silver particles, gold particles, a mixture of silver and tin particles, etc.

As depicted in FIG. 4, electrical traces **154** and **155** are patterned from a single electrically conductive metal layer disposed over a thermally conductive metal core layer **152**. Dielectric layer **153** electrically isolates electrical traces **154** and **155** from thermally conductive metal core layer **152**. As depicted in FIG. 4, LED based illumination source **156** includes top facing electrical interconnect pads **157** and **158**. Electrical interconnect pad **157** is electrically coupled to electrical trace **154** via wire bond **159**. Similarly, electrical interconnect pad **158** is electrically coupled to electrical trace **155** via wire bond **160**. Electrical traces **154** and **155** are electrically isolated from one another and an electrical current is driven through LED based illumination source **156** via a voltage difference maintained between electrical traces **154** and **155**. The electrical current induces light emission **161** from LED based illumination source **156**.

The MC-PCB mechanically and electrically interfaces LED based illumination source **156** with the application environment in which LED based illumination source **156** is installed. The electrically isolated backside surface of LED based illumination source **156** does not participate in the electrical circuit driving current through LED based illumination source **156**. Thus, the coupling of the backside surface of LED based illumination source **156** to metal core layer **152** is optimized for thermal performance without consideration of electrical performance. For example, the thermally conductive paste **162** is selected to be as thermally conductive as possible without regard for electrical resistance because no particular electrical isolation or conductivity is required at the interface between the LED based illumination source **156** and the metal core layer **152**.

In a similar manner, electrically isolated LED based illumination sources may be electrically coupled in series using wire bonds with high spatial density. FIG. 5 is a simplified diagram illustrative of a cross-sectional view of multiple electrically isolated LED based illumination sources **156A-C** assembled onto a single layer MC-PCB and electrically coupled in series in a wire bond configuration. Like numbered elements depicted in FIG. 4 are analogous. As depicted in FIG. 5, LED based illumination sources **156A-C** each include top facing electrical interconnect pads electrically coupled to an anode and a cathode, respectively, or each corresponding LED based illumination source.

Electrical interconnect pad **157A** of LED based illumination source **156A** is electrically coupled to electrical trace **154** via wire bond **159**. Electrical interconnect pad **157B** of LED based illumination source **156A** is electrically coupled to electrical interconnect pad **162A** of LED based illumination source **156B** via wire bond **163**. Electrical interconnect pad **162B** of LED based illumination source **156B** is electrically coupled to electrical interconnect pad **158A** of LED based illumination source **156C** via wire bond **164**. Electrical interconnect pad **158B** of LED based illumination source **156C** is electrically coupled to electrical trace **155** via wire bond **160**. Electrical traces **154** and **155** are electrically isolated from one another and an electrical current is driven through LED based illumination sources **156A-C** via a voltage difference maintained between electrical traces **154** and **155**. The electrical current induces light emission **161** from LED based illumination sources **156A-C**.

As illustrated in FIG. 5, electrically isolated LEDs flow current through top facing terminals rather than the backside surface of the LED die. As a result, multiple bare LED die may be mounted to a common MC-PCB and wired in series using wire bonds on top of the LED die. In high optical

power applications, it is advantageous to couple multiple bare LED die in series to minimize the current that must be generated by a current driver to drive multiple LED die. In one example, each of bare LED die **156A-C** has a current rating of four Ampere. When coupled in series, a single current driver rated at four ampere is capable of driving LED die **156A-C**. When coupled in parallel, a single current driver rated at twelve Ampere is required to drive LED die **156A-C**, or alternatively, three current drivers, each rated at four Ampere is required to drive LED die **156A-C**. A single twelve Ampere driver is significantly more expensive and is significantly less electrically efficient compared to a four Ampere driver. Similarly, the use of three different four Ampere drivers is significantly more expensive and complex compared to the use of a single four Ampere driver.

Furthermore, it is advantageous to couple multiple bare LED die in series to maximize the spatial density of LED die mounted to a MC-PCB. This minimizes the illumination source etendue and simplifies the optics subsystem required to focus the illumination light on the workpiece. In other words, it is desirable to mount multiple LED die onto an MC-PCB as close together as possible to minimize the size and complexity of the optics subsystem, e.g., optics subsystem **104** depicted in FIG. **1**, required to collect and focus illumination light onto the workpiece under process. When coupled in series, it is straightforward to mount LED die in close proximity. However, when coupled in parallel, an electrical trace must be located between LED die to address the cathode or anode of each LED die. As a result, there are significant limitations to how closely LED die can be mounted to a MC-PCB, when the LED die are coupled in parallel.

Although, FIG. **5** depicts three LED die wired in series, and FIGS. **2**, **3**, **7**, and **10** depict four LED die wired in series, in general, a hybrid LED and laser diode based illumination device may include any integer number of LED die greater than one wired in series.

FIG. **6** is a simplified diagram illustrative of a cross-sectional view of an electrically isolated LED based illumination source **156** assembled onto a single layer MC-PCB with a pedestal structure **172**. Like numbered elements depicted in FIG. **4** are analogous. As depicted in FIG. **6**, the electrically isolated backside of LED based illumination source **156** is disposed over thermally conductive metal core layer **152**. In addition, a thermally conductive pedestal structure **172** is disposed between LED based illumination source **156** and MC-PCB **152** to adjust the vertical location of LED based illumination source **156** relative to electrical traces **154** and **155**. As depicted in FIG. **6**, heat flow **165** carries thermal energy away from LED based illumination source **156**, through pedestal **172**, into thermally conductive metal core layer **152**. A thin layer of thermally conductive paste **162** is located between LED based illumination source **156** and pedestal **172** and between pedestal **172** and thermally conductive metal core layer **152**. Pedestal structure **172** is fabricated from a thermally conductive material or combination of materials, e.g., copper, tin, silver, gold, etc.

FIG. **7** depicts a simplified illustration of a hybrid LED and laser diode based illumination device **131** including an array **136** of electrically isolated LEDs and a laser diode **137** mounted to a metal core printed circuit board (MC-PCB) **139**. Like numbered elements depicted in FIG. **2** are analogous.

As depicted in FIG. **7**, electrical traces **132-135** are patterned from a single electrically conductive metal layer disposed over a thermally conductive metal core layer. Terminal connector **118** includes four terminals **118A-D**,

electrically coupled to electrical traces **132-135**, respectively. In this manner, electrical terminal **118A** is electrically coupled to one voltage node of LED array **136**, electrical terminal **118B** is electrically coupled to another voltage node of LED array **136**, electrical terminal **118C** is electrically coupled to one voltage node of laser diode **137**, and electrical terminal **118D** is electrically coupled to another voltage node of laser diode **137**.

As depicted in FIG. **7**, LED array **136** is subdivided into two subarrays of bare LED die **136A** and **136B**. The two subarrays are electrically coupled in series via electrical trace **138** patterned from the single electrically conductive metal layer of MC-PCB **139**. The electrical trace **138** is shaped such that laser diode **137** is located between the two subarrays of bare LED die **136A** and **136B**. By locating the laser diode source within the spatial envelope B of the LED array **136**, a particularly compact, high optical power, hybrid LED and laser diode based illumination device is realized. By locating the laser diode source within the spatial envelope B of the LED array, the illumination source etendue of the hybrid LED and laser diode based illumination device may be further reduced compared to a design where the laser diode source is located outside the spatial envelope of the LED array.

FIG. **8** depicts an illustration of a hybrid LED and laser diode based illumination device **181** in another embodiment. As depicted in FIG. **8**, a plurality of electrically isolated LED based illumination sources, e.g., bare LED die, are disposed around one or more electrically isolated laser diode based illumination sources. In particular, FIG. **8** depicts bare LED die **182A-J** disposed around laser diode based illumination source **183**.

FIG. **9** depicts an illustration of a hybrid LED and laser diode based illumination device **191** in another embodiment. As depicted in FIG. **9**, one or more electrically isolated laser diode based illumination sources are disposed in a pattern interspersed among the plurality of electrically isolated LED based illumination sources. In particular, FIG. **9** depicts patterns of laser diode based illumination sources **193A** and **193B** interspersed among bare LED die **192A-C**.

In embodiments employing a multiple layer MC-PCB, multiple electrically conductive metal layers are disposed over a thermally conductive metal core layer in a stacked configuration. In these embodiments, a dielectric layer is disposed between each of the electrically conductive metal layers and the thermally conductive metal core layer in the stack.

In some embodiments, a hybrid LED and laser diode based illumination device includes a multiple layer MC-PCB. In some of these embodiments, the electrical traces electrically coupled to one of more laser diode based illumination sources are patterned from a different electrically conductive metal layer than the electrical traces electrically coupled to one or more LED based illumination sources of the hybrid LED and laser diode based illumination device.

In some embodiments, an LED based illumination source is a group III nitride semiconductor structure, including, by way of non-limiting example, an n layer, a light emitting layer, and a p layer. In various examples, the group III nitride semiconductor structure is AlGaIn-based, GaN-based, or InAlGaIn-based. In general, a group III nitride semiconductor structure is configured to emit wavelengths in the ultraviolet spectral band, e.g., ultraviolet light in a range between 365 nanometers and 405 nanometers, the visible spectral band, e.g., blue light having wavelength greater than 405 nanometers, or both.

In some embodiments, an LED based illumination source of a hybrid LED and laser diode based illumination device is a packaged LED illumination source.

In some other embodiments, an LED based illumination source of a hybrid LED and laser diode based illumination device is a bare LED die. In some of these embodiments, each of the LED based illumination sources of a hybrid LED and laser diode based illumination device are bare LED die.

An emission area of each of the plurality of LED based illumination sources of a hybrid LED and laser diode based illumination device is characterized by a length dimension and a width dimension orthogonal to the length dimension. In some embodiments, the length and width dimension of an LED based illumination source of a hybrid LED and laser diode based illumination device is less than 1.5 millimeters by 1.5 millimeters. In some embodiments, the length and width dimension of each LED based illumination source of a hybrid LED and laser diode based illumination device is less than 50 micrometers by 50 micrometers.

In some embodiments, the one or more laser diode based illumination sources of a hybrid LED and laser diode based illumination device are packaged devices. In some of these embodiments, the packages are surface mounted, e.g., a ceramic laser package, a surface mount TO package, etc. In some other embodiments, a laser diode based illumination source of a hybrid LED and laser diode based illumination device is a bare laser diode die. In preferred embodiments, the one or more laser diode based illumination sources are electrically isolated. However, in some embodiments, one or more laser diode based illumination sources are not electrically isolated. In these embodiments, thermally conductive metal core layer of the metal core printed circuit board (MC-PCB) is part of the electrical circuit driving current through the one or more laser diode devices, e.g., the thermally conductive metal core layer is a ground plane in an electrical circuit including the one or more laser diode devices.

In some embodiments, one or more LED based illumination sources of a hybrid LED and laser diode based illumination device emits light at a different peak wavelength than other LED based illumination sources of the hybrid LED and laser diode based illumination device.

In some embodiments, one or more LED based illumination sources of a hybrid LED and laser diode based illumination device emits light at a different peak wavelength than one or more laser diode based illumination sources of the hybrid LED and laser diode based illumination device.

In some examples, different LED based illumination sources of a hybrid LED and laser diode based illumination device emits light with a peak wavelength of 385 nanometers, 365 nanometers, 940 nanometers, or any other suitable peak wavelength. In some of these examples, a LED based illumination source of a hybrid LED and laser diode based illumination device emits light with a peak wavelength in the infrared portion of the electromagnetic spectrum.

In some examples, a laser diode based illumination source of a hybrid LED and laser diode based illumination device emits light with a peak wavelength of 405 nanometers. In some other examples, a laser diode based illumination source of a hybrid LED and laser diode based illumination device emits light in the infrared spectral range, e.g., 940 nanometers, to initiate thermal curing of various materials. However, in general, a laser diode based illumination source of a hybrid LED and laser diode based illumination device may be configured to emit light at any suitable peak wavelength.

In one example, a hybrid LED and laser diode based illumination device includes LED based illumination sources emitting at a peak wavelength at or between 365 nanometers and 385 nanometers and a laser diode based illumination source emitting at a peak wavelength of 405 nanometers. This exemplary hybrid LED and laser diode based illumination device is capable of curing solder mask inks employed in many different applications, e.g., cell phone components, automotive components, etc. For example, such a hybrid LED and laser diode based illumination device may be employed to cure white solder mask ink, which cures at 405 and 415 nanometers, green solder mask ink, which cures at 365 nanometers and 385 nanometers, and black solder mask ink, which cures at 405 nanometers.

In some embodiments, both LED based illumination sources and laser diode based illumination sources of a hybrid LED and laser diode based illumination device are operated simultaneously. As a result the illumination output of the hybrid LED and laser diode based illumination device includes multiple wavelengths. In one example, a current driver coupled to terminals **118A** and **118B** depicted in FIG. **2** supplies current to LED array **116** at the same time another current driver coupled to terminals **118C** and **118D** supplies current to laser diode illumination source **117**.

However, in some other embodiments, the LED based illumination sources and laser diode based illumination sources of a hybrid LED and laser diode based illumination device are operated separately. For example, it is known that some types of green ink are optimally cured at peak wavelengths of 365 nanometers and 385 nanometers, but are damaged by wavelengths at 405 nanometers. In this example, it is desirable to illuminate the green ink with the illumination output of the LED based illumination sources and not the laser diode based illumination source.

In a further aspect, different sets of LEDs and one or more laser diodes are elements of different electrical circuits that are independently controlled to generate illumination with different spectral characteristics. In this manner, the illumination characteristics of the hybrid LED and laser diode based illumination device are adapted to applications that require different combinations of illumination wavelengths. In some examples, different combinations of illumination sources of the hybrid LED and laser diode based illumination device are employed using the same optical subsystem to perform industrial curing of a wide range of materials. In some embodiments, LEDs and LDs are employed at ultraviolet wavelengths at relatively high optical power to cure different materials, e.g., adhesives, etc.

FIG. **10** depicts a simplified illustration of a hybrid LED and laser diode based illumination device **141** including an array **146** of electrically isolated LEDs and a laser diode **147** mounted to a metal core printed circuit board (MC-PCB) **149**. Like numbered elements depicted in FIGS. **2** and **7** are analogous.

As depicted in FIG. **10**, LED array **146** includes three different LED sub-arrays **146A-C**. Electrical traces **142**, **143A-C**, **144**, and **145**, are patterned from one or more electrically conductive metal layers disposed over a thermally conductive metal core layer. Terminal connector **148** includes six terminals **148A-F**, electrically coupled to electrical traces **142**, **143A-C**, **144**, and **145**, respectively. In this manner, electrical terminal **148A** is electrically coupled to a voltage node of the LED sub-arrays **146A-C**, electrical terminals **148B-D** are electrically coupled to the opposite voltage nodes of LED sub-arrays **146A-C**, respectively, electrical terminal **148E** is electrically coupled to a voltage

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node of laser diode **147**, and electrical terminal **148F** is electrically coupled to the opposite voltage node of laser diode **147**.

As depicted in FIG. **10**, LED array **146** is subdivided into three subarrays **146A-C** of bare LED die. Each of the three sub-arrays are elements of different electrical circuits. In the embodiment depicted in FIG. **10**, one voltage node of the three subarrays share a common ground terminal, i.e., terminal **148A**, but the opposite voltage node of each of the three subarrays is electrically coupled to three different current drivers. In this embodiment, one current driver is electrically coupled between terminals **148A** and **148B**, a second current driver is electrically coupled between terminals **148A** and **148C**, and a third current driver is electrically coupled between terminals **148A** and **148D**. In addition, a fourth current driver is electrically coupled between terminals **148E** and **148F**.

Each different current driver is independently controlled, and thus the current flow through each of LED sub-arrays **146A-C** and laser diode **147** is independently controlled. Each different current driver addresses an independent illumination "channel". The embodiment depicted in FIG. **10** includes four independent illumination channels. However, in general, a hybrid LED and laser diode based illumination device may include any number of independent channels. In general, an independent channel includes one illumination source or multiple illumination sources in series, or multiple illumination sources coupled in series and parallel.

In some embodiments, the peak wavelength of at least one illumination channel is different. In some embodiments, the peak wavelength of each illumination channel is different. In many embodiments, a channel includes 3-6 bare LED die are electrically coupled in series.

FIG. **11** illustrates a method **200** suitable for implementation by a hybrid LED and laser diode based illumination device of the present invention. While the following description is presented in the context of hybrid LED and laser diode based illumination devices **111**, **131**, **141**, **181**, and **191**, it is recognized herein that the particular structural aspects of hybrid LED and laser diode based illumination devices **111**, **131**, **141**, **181**, and **191** do not represent limitations and should be interpreted as illustrative only.

In block **201**, an amount of illumination light is generated from a hybrid illumination source. The hybrid illumination source includes a metal core printed circuit board (MC-PCB) having a thermally conductive metal core layer, one or more patterned, electrically conductive metal layers disposed over one or more surfaces of the thermally conductive metal core layer, and one or more dielectric layers disposed between the one or more patterned, electrically conductive metal layers, disposed between the thermally conductive metal core layer and a patterned, electrically conductive metal layer of the one or more patterned, electrically conductive layers, or both. The one or more dielectric layers are electrically insulative. The hybrid illumination source also includes a plurality of electrically isolated LED based illumination sources. Each of the plurality of electrically isolated LED based illumination sources is mechanically coupled to the thermally conductive metal core layer. The hybrid illumination source also includes one or more laser diode based illumination sources. Each of the one or more laser diode based illumination sources is mechanically coupled to the thermally conductive metal core layer.

In block **202**, the amount of light emitted from the hybrid illumination source is directed to a workpiece under process by an optics subsystem. An etendue of each of the one or more laser diode based illumination sources and each of the

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plurality of electrically isolated LED based illumination sources overlap over an area at a plane perpendicular to an optical path through the optics subsystem.

Although certain specific embodiments are described above for instructional purposes, the teachings of this patent document have general applicability and are not limited to the specific embodiments described above. Accordingly, various modifications, adaptations, and combinations of various features of the described embodiments can be practiced without departing from the scope of the invention as set forth in the claims.

What is claimed is:

1. A hybrid illumination device comprising:

a metal core printed circuit board (MC-PCB) having a thermally conductive metal core layer, one or more patterned, electrically conductive metal layers disposed over the thermally conductive metal core layer, and one or more dielectric layers disposed between the one or more patterned, electrically conductive metal layers, disposed between the thermally conductive metal core layer and a patterned, electrically conductive metal layer of the one or more patterned, electrically conductive layers, or both, wherein the one or more dielectric layers are electrically insulative;

a plurality of electrically isolated LED based illumination sources, each of the plurality of electrically isolated LED based illumination sources having a surface area disposed over the thermally conductive metal core layer, wherein an amount of heat generated by each of the plurality of electrically isolated LED based illumination sources flows through the surface area to the thermally conductive metal core layer, wherein the surface area is not electrically active, wherein at least one of the plurality of electrically isolated LED based illumination sources is electrically coupled to a first of the one or more patterned, electrically conductive metal layers; and

one or more laser diode based illumination sources, each of the plurality of laser diode based illumination sources disposed over the thermally conductive metal core layer, wherein at least one of the one or more laser diode based illumination sources is electrically coupled to the first of the one or more patterned, electrically conductive metal layers or a second of the one or more patterned, electrically conductive metal layers.

2. The hybrid illumination source of claim 1, wherein the plurality of electrically isolated LED based illumination sources are disposed around the one or more laser diode based illumination sources.

3. The hybrid illumination source of claim 1, wherein the one or more laser diode based illumination sources are disposed in a pattern interspersed among the plurality of electrically isolated LED based illumination sources.

4. The hybrid illumination source of claim 1, wherein the one or more laser diode based illumination sources are electrically isolated.

5. The hybrid illumination source of claim 1, wherein each of the plurality of LED based illumination sources are bare die.

6. The hybrid illumination source of claim 1, wherein an emission area of each of the plurality of LED based illumination sources is characterized by a length dimension and a width dimension orthogonal to the length dimension, and wherein the length dimension and the width dimension are less than 50 micrometers.

7. The hybrid illumination source of claim 1, wherein a first of the plurality of electrically isolated LED based

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illumination sources is electrically coupled to a first conductive trace of a first of the one or more patterned, electrically conductive metal layers, wherein a second of the plurality of electrically isolated LED based illumination sources is electrically coupled to a second conductive trace of the first patterned, electrically conductive metal layer, wherein the first and second electrically isolated LED based illumination sources are electrically coupled in series, and wherein the first and second conductive traces are electrically isolated from one another.

8. The hybrid illumination source of claim 7, wherein a first of the one or more laser diode based illumination sources is electrically coupled to a first conductive trace of the second of the one or more patterned, electrically conductive metal layers or a third conductive trace of the first of the one or more patterned, electrically conductive metal layers.

9. The hybrid illumination source of claim 7, wherein a third of the plurality of electrically isolated LED based illumination sources is electrically coupled to a third conductive trace of the first patterned, electrically conductive metal layer, wherein a fourth of the plurality of electrically isolated LED based illumination sources is electrically coupled to the second conductive trace or a fourth conductive trace of the first patterned, electrically conductive metal layer, wherein the third and fourth electrically isolated LED based illumination sources are electrically coupled in series, and wherein the third and conductive trace is electrically isolated from the second conductive trace.

10. The hybrid illumination source of claim 9, wherein a peak wavelength of light emitted from the first and second electrically isolated LED based illumination sources is different from a peak wavelength of light emitted from the third and fourth electrically isolated LED based illumination sources.

11. The hybrid illumination source of claim 10, wherein a current flow through the first and second electrically isolated LED based illumination sources is controlled independently from a current flow through the third and fourth electrically isolated LED based illumination sources.

12. The hybrid illumination source of claim 8, wherein a peak wavelength of light emitted from the first and second electrically isolated LED based illumination sources is different from a peak wavelength of light emitted from the first of the one or more laser diode based illumination sources.

13. The hybrid illumination source of claim 12, wherein a current flow through the first and second electrically isolated LED based illumination sources is controlled independently from a current flow through the first of the one or more laser diode based illumination sources.

14. A hybrid illumination system comprising:

a hybrid illumination source comprising:

a metal core printed circuit board (MC-PCB) having a thermally conductive metal core layer, one or more patterned, electrically conductive metal layers disposed over one or more surfaces of the thermally conductive metal core layer, and one or more dielectric layers disposed between the one or more patterned, electrically conductive metal layers, disposed between the thermally conductive metal core layer and a patterned, electrically conductive metal layer of the one or more patterned, electrically conductive layers, or both, wherein the one or more dielectric layers are electrically insulative;

a plurality of electrically isolated LED based illumination sources, each of the plurality of electrically

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isolated LED based illumination sources mechanically coupled to the thermally conductive metal core layer; and

one or more laser diode based illumination sources, each of the plurality of laser diode based illumination sources mechanically coupled to the thermally conductive metal core layer; and

an optics subsystem configured to direct light emitted from the hybrid illumination source to a workpiece under process, wherein an etendue of each of the one or more laser diode based illumination sources and each of the plurality of electrically isolated LED based illumination sources overlap over an area at a plane perpendicular to an optical path through the optics subsystem.

15. The hybrid illumination system of claim 14, wherein a first set of the plurality of electrically isolated LED based illumination sources are electrically coupled in series, wherein a second set of the plurality of electrically isolated LED based illumination sources are electrically coupled in series, and wherein the first set, the second set and the laser diode based illumination source are elements of independent electrical circuits.

16. The hybrid illumination system of claim 15, wherein a peak wavelength of light emitted from the first set of electrically isolated LED based illumination sources is different from a peak wavelength of light emitted from the one or more laser diode based illumination sources, and wherein a current flow through the first set of electrically isolated LED based illumination sources is controlled independently from a current flow through the one or more laser diode based illumination sources.

17. The hybrid illumination system of claim 15, wherein a peak wavelength of light emitted from the first set of electrically isolated LED based illumination sources is different from a peak wavelength of light emitted from the second set of electrically isolated LED based illumination sources, and wherein a current flow through the first set of electrically isolated LED based illumination sources is controlled independently from a current flow through the second set of electrically isolated LED based illumination sources.

18. A method comprising:

generating an amount of illumination light from a hybrid illumination source, the hybrid illumination source including a metal core printed circuit board (MC-PCB) having a thermally conductive metal core layer, one or more patterned, electrically conductive metal layers disposed over one or more surfaces of the thermally conductive metal core layer, and one or more dielectric layers disposed between the one or more patterned, electrically conductive metal layers, disposed between the thermally conductive metal core layer and a patterned, electrically conductive metal layer of the one or more patterned, electrically conductive layers, or both, wherein the one or more dielectric layers are electrically insulative; a plurality of electrically isolated LED based illumination sources, each of the plurality of electrically isolated LED based illumination sources mechanically coupled to the thermally conductive metal core layer; and one or more laser diode based illumination sources mechanically coupled to the thermally conductive metal core layer; and

directing the amount of light emitted from the hybrid illumination source to a workpiece under process by an optics subsystem, wherein an etendue of each of the one or more laser diode based illumination sources and

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each of the plurality of electrically isolated LED based illumination sources overlap over an area at a plane perpendicular to an optical path through the optics subsystem.

19. The method of claim 18, wherein a set of the plurality 5 of electrically isolated LED based illumination sources are electrically coupled in series, and wherein the first set and the one or more laser diode based illumination sources are elements of independent electrical circuits.

20. The method of claim 19, wherein a peak wavelength 10 of light emitted from the set of electrically isolated LED based illumination sources is different from a peak wavelength of light emitted from the one or more laser diode based illumination sources, and wherein a current flow 15 through the set of electrically isolated LED based illumination sources is controlled independently from a current flow through the one or more laser diode based illumination sources.

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